

Product Requirements Document: Benefit Card Wallet Fallback Logic

1. Overview

1.1 Document Purpose

This PRD defines the requirements for implementing a fallback mechanism across multiple benefit wallets for benefit card transactions, enabling seamless transaction approval when the primary wallet has insufficient funds.

1.2 Background

Currently, benefit cards are linked to multiple wallets (food/meal, telecom, fuel, rewards, and general account). Each benefit wallet (except rewards and general) is mapped to specific MCC codes. When a transaction is initiated, the system checks only the primary mapped wallet, rejecting the transaction if insufficient funds exist—even when other wallets contain adequate balance.

1.3 Problem Statement

- **Current Behavior:** Transactions are declined if the primary mapped wallet has insufficient balance, regardless of available balance in other wallets
 - **Business Impact:** Poor customer experience and reduced card utilization despite cardholders having sufficient total balance across all wallets
 - **Customer Pain Point:** Cardholders are forced to manually manage wallet balances or face transaction declines unnecessarily
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2. Goals & Objectives

2.1 Primary Goals

- Implement a hierarchical fallback mechanism across benefit wallets to improve transaction approval rates
- Enhance cardholder experience by intelligently utilizing available balances across wallets
- Maintain clear wallet prioritization logic based on transaction MCC codes

2.2 Success Metrics

- Increase in transaction approval rate for benefit card holders

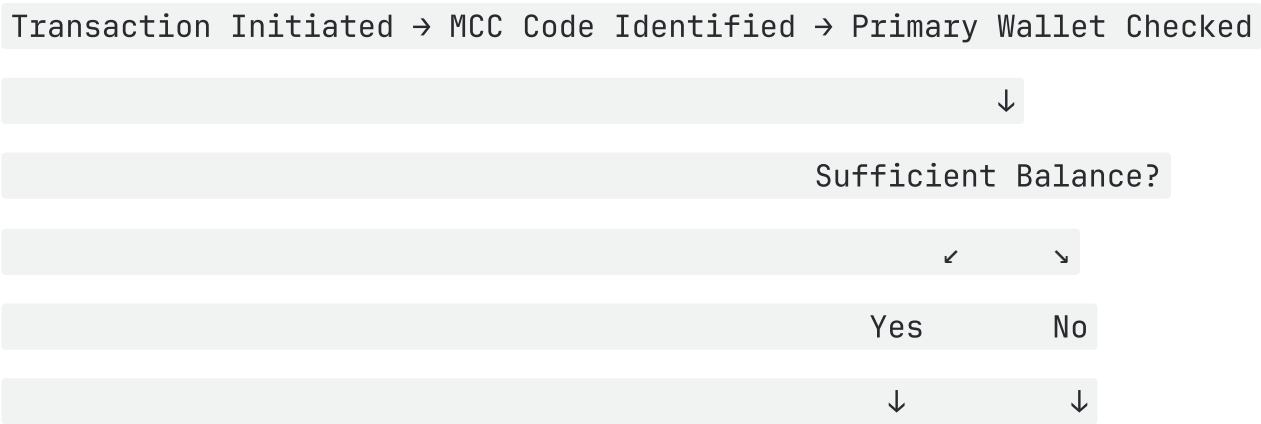
- Reduction in declined transactions due to insufficient funds in primary wallet
- Improved customer satisfaction scores related to card usage
- Decreased customer support inquiries regarding declined transactions

3. Wallet Structure & Current Behavior

3.1 Wallet Types

Wallet Type	MCC Mapping	Current Behavior
Food/Meal Wallet	Fixed MCC list (food/meal related)	Checked for food/meal MCC transactions
Telecom Wallet	Fixed MCC list (telecom related)	Checked for telecom MCC transactions
Fuel Wallet	Fixed MCC list (fuel related)	Checked for fuel MCC transactions
Rewards Wallet	All MCCs allowed (no fixed mapping)	Checked for unmapped MCCs; serves as secondary fallback
General Wallet	All MCCs allowed (no fixed mapping)	Main Tide account; ultimate fallback

3.2 Current Transaction Flow



Approve Decline

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3.3 Current Limitations

1. ****Benefit-Specific Transactions****: If food/meal MCC transaction has insufficient balance in food wallet → transaction declined (even if rewards/general wallet has balance)
2. ****Unmapped MCC Transactions****: If transaction MCC is not mapped to any benefit wallet → checks general wallet only → if insufficient, transaction declined (rewards wallet not checked)
3. ****No Cross-Wallet Utilization****: No mechanism to leverage available balance across multiple wallets

4. Requirements

4.1 Functional Requirements

4.1.1 Fallback Hierarchy (Priority Order)

****For Transactions with Benefit Wallet MCC Mapping**** (Food/Meal, Telecom, Fuel):

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1. Check Benefit-Specific Wallet (e.g., Food Wallet for food MCC)
 ↓ (if insufficient)
2. Check Rewards Wallet
 ↓ (if insufficient)

3. Check General Wallet

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****For Transactions without Benefit Wallet MCC Mapping**:**

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1. Check Rewards Wallet

↓ (if insufficient)

2. Check General Wallet

...

4.1.2 Full Balance Requirement

- ****FR-001****: The transaction amount must be fully available in a single wallet; partial deductions across multiple wallets are NOT supported
- ****FR-002****: Each wallet in the fallback chain must be checked for the complete transaction amount
- ****FR-003****: Transaction is approved from the first wallet in the hierarchy that contains sufficient funds

4.1.3 Transaction Processing Logic

****FR-004: MCC-to-Wallet Mapping****

- System must identify the transaction MCC code from the acquirer
- System must determine if MCC maps to a specific benefit wallet (food/meal, telecom, fuel)
- If no mapping exists, treat as unmapped MCC transaction

****FR-005: Wallet Balance Verification****

- For each wallet in the fallback hierarchy:

- Check if available balance $\geq$ transaction amount
- If yes: proceed with authorization from that wallet
- If no: move to next wallet in hierarchy
- If all wallets have insufficient balance: decline transaction

****FR-006: Authorization Response****

- Upon successful balance verification, deduct full transaction amount from the identified wallet
- Return approval with wallet identifier used for the transaction
- If all fallbacks fail, return decline with appropriate reason code

4.2 Non-Functional Requirements

****NFR-001: Performance****

- Wallet balance checks must complete within 2/5 secs to meet authorization timeframe requirements (Tide is allowed to take maximum of 1 sec)
- Fallback logic should add no more than XXms to existing authorization latency

****NFR-002: Reliability****

- Fallback mechanism must have 99.95% success rate in executing the correct wallet checks
- System must handle concurrent transaction scenarios without race conditions on wallet balances

****NFR-003: Auditability****

- All wallet checks during fallback process must be logged

- Transaction records must clearly indicate which wallet was ultimately used for authorization
- Audit trail must capture the complete fallback chain evaluation

****NFR-004: Scalability****

- Solution must support the fallback logic for peak transaction volumes (define specific TPS based on your system)
- Database queries for wallet balance checks must be optimized for concurrent access

5. Detailed Use Cases & Scenarios

5.1 Scenario 1: Food Transaction with Food Wallet Insufficient Balance

****Setup:****

- Food Wallet: \$10
- Rewards Wallet: \$50
- General Wallet: \$100
- Transaction: \$25 (Food MCC)

****Expected Flow:****

1. System identifies Food MCC → checks Food Wallet first
2. Food Wallet has \$10 (insufficient)
3. System checks Rewards Wallet
4. Rewards Wallet has \$50 (sufficient)

5. ****Transaction approved from Rewards Wallet****
6. Post-transaction balances: Food: \$10, Rewards: \$25, General: \$100

5.2 Scenario 2: Food Transaction Requires General Wallet Fallback

****Setup:****

- Food Wallet: \$5
- Rewards Wallet: \$15
- General Wallet: \$100
- Transaction: \$30 (Food MCC)

****Expected Flow:****

1. System identifies Food MCC → checks Food Wallet first
2. Food Wallet has \$5 (insufficient)
3. System checks Rewards Wallet
4. Rewards Wallet has \$15 (insufficient)
5. System checks General Wallet
6. General Wallet has \$100 (sufficient)
7. ****Transaction approved from General Wallet****
8. Post-transaction balances: Food: \$5, Rewards: \$15, General: \$70

5.3 Scenario 3: Unmapped MCC Transaction

****Setup:****

- Rewards Wallet: \$20
- General Wallet: \$150
- Transaction: \$50 (Entertainment MCC - not mapped to any benefit wallet)

****Expected Flow:****

1. System identifies unmapped MCC
2. Skips benefit wallet checks
3. System checks Rewards Wallet first
4. Rewards Wallet has \$20 (insufficient)
5. System checks General Wallet
6. General Wallet has \$150 (sufficient)
7. ****Transaction approved from General Wallet****
8. Post-transaction balances: Rewards: \$20, General: \$100

5.4 Scenario 4: All Wallets Insufficient Balance

****Setup:****

- Food Wallet: \$10
- Rewards Wallet: \$15
- General Wallet: \$20
- Transaction: \$50 (Food MCC)

****Expected Flow:****

1. System checks Food Wallet → \$10 (insufficient)
2. System checks Rewards Wallet → \$15 (insufficient)
3. System checks General Wallet → \$20 (insufficient)
4. ****Transaction declined - Insufficient funds****
5. Decline reason: "Insufficient balance across all available wallets"

Define wallets at SP - does not offer reward benefits

Linked Member - does not have rewards wallet

Open question - which wallet will the decline txn/reason be shown for?

5.5 Scenario 5: Multiple Benefit Wallet Types

Setup:

- Fuel Wallet: \$30
- Telecom Wallet: \$25
- Rewards Wallet: \$60
- General Wallet: \$200
- Transaction: \$40 (Fuel MCC)

Expected Flow:

1. System identifies Fuel MCC → checks Fuel Wallet first
2. Fuel Wallet has \$30 (insufficient)
3. System checks Rewards Wallet
4. Rewards Wallet has \$60 (sufficient)
5. **Transaction approved from Rewards Wallet**
6. Post-transaction balances: Fuel: \$30, Telecom: \$25, Rewards: \$20, General: \$200